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Lightweight self-hosted AI chat web app. An Open WebUI alternative with **under 100 MB RAM**. No Docker. No build step. One Node.js process + one SQLite file.

Stack	Node.js + Express 5 + better-sqlite3 + vanilla JS
AI Backend	9Router (<code>http://localhost:20128/v1</code> , OpenAI-compatible)
Providers	AG, Groq, BAI, OpenRouter, Nvidia, Cline — 497+ models
Frontend	<code>index.html</code> + <code>style.css</code> + <code>app.js</code> , zero framework, dark theme
Language	English (default) / Indonesian — runtime toggle
Version	v1.0-beta
License	Private / personal use

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Features

Chat

Feature	Description
Real-time streaming	SSE relay, word-by-word from 9Router
Multi-session history	Per-user isolation in SQLite

Feature	Description
Chat memory	Token-budget context window (default 1600 tokens ≈ 15–30 msgs, oldest auto-compressed) + rolling background summary of older turns
Cross-chat memory	Auto-extracted personal facts per user (max 20), injected into every chat; editable in Settings → Memory
Token saver	<code>max_tokens</code> reply cap (default 1024), history budget, per-chat Lean ⚡ toggle (answers without history)
Per-reply usage badge	prompt / completion / total tokens shown under each AI reply
Rename / Delete chats	Via sidebar hover buttons; rename uses custom in-app dialog (bilingual, no native prompt)
Pin / Archive / Tag / Fork	Chat "...": pin to top, archive to a collapsible "Archived (n)" section, inline #tag, fork = branch copy of the conversation ((fork) suffix)
Chat search	Instant client-side filter by title (sidebar)
Markdown + code blocks	With streaming auto-close fence fix
Copy & Retry	Per-message actions
Smart chat title	After the first reply a short 3–6 word title is generated in the background (never overwrites a manual rename)
Edit & regenerate	Pencil on any user message → inline edit → save truncates everything after it; ↺ button regenerates from the last user message
Code highlight + copy	fenced blocks get language colouring (bundled hljs, no CDN) + per-block copy button
Stop generation	Abort mid-stream
Provider timeout	Admin-configurable 30–600 s (default 120 s), applies to all upstream calls
Typing indicator	Animated dots while waiting
Temporary Chat	Zero history saved (endpoint: <code>/api/temp-chat</code>); honours Lean mode (skip history + memory = zero context)
Realtime quota UI	Sidebar quota + admin dashboard refresh automatically after every reply (dashboard auto-polls every 5 s while open)

Users & Access

Feature	Description
Login	script per-user salt (<code>script1:salt:hash</code>), timing-safe compare; legacy static-salt hashes auto-upgrade on login
Session purge	Expired sessions deleted on every login
Roles	<code>admin</code> (full control) and <code>user</code> (chat only)
Daily quota	Per-user limit (default 50/day), admin unlimited; quota bar shown in sidebar with model chip
User management	Admin: add / delete / reset password / activate / deactivate (self-deletion & self-deactivation blocked server-side)
Instant session revoke	Deactivate or password reset kills sessions immediately

Feature	Description
Quota edit & reset	Admin edits any user's daily quota (1–999999) and resets one or all users' usage
Reset password (admin)	Custom in-app modal with validation (min 4 chars, clear on short) — no native <code>prompt()</code> anywhere
Deactivate/Activate/Delete guard	Open WebUI-style warn modals with user name + consequence text (red danger for deactivate/delete, blue/info for activate); actions only run after confirm

Guest Mode (no login)

Feature	Description
Try as guest	Button under Login on the login screen (only shown when guest mode is enabled + gateway configured); clicks straight into the chat dashboard
Configurable limits	Admin sets max messages per guest (1–200, default 10) and session time limit (1–60 min, default 5)
Live guest chip	Sidebar shows <code>Guest — n/max messages · mm:ss left</code> (turns red under 60 s)
Login popup on exhaustion	When chats or time run out, an in-app modal asks to sign in; "Continue as guest" dismisses it, "Sign in now" goes to login
Server-enforced	Limits checked before the AI call; rejected sends roll back their bubbles (nothing stored); expired sessions can't post to any account endpoint (<code>ERR_GUEST_NOPE</code>)
Session persistence	Cookie <code>gtoken</code> (HttpOnly, 24 h); chat history kept during the session, survives reload; re-clicking guest mints a fresh session
Clean stats	Guest messages never touch the <code>usage</code> table — admin statistics stay per-real-user
Auto-delete (nightly purge)	Scheduler wipes ALL guest data — sessions, chats, messages — once per day at an admin-configurable local hour (default 00:00, <code>guest_purge_hour</code>); plus a manual "Purge guest chats now" button in Admin → Usage Statistics

Chat Experience

Feature	Description
Smart title	After the first reply, a short 3–6-word title is generated in the background (never overwrites manual renames; guests excluded)
Edit message	Pencil on user bubbles → inline textarea → save = rewrite + truncate everything after, resend automatically
Regenerate	↻ on AI bubbles truncates from the last user message and resends
Code highlighting	Fenced blocks coloured via bundled highlight.js (v11.9, served from <code>/vendor/</code> , no CDN) + per-block copy button; theme-aware (github-dark/github swap with the UI theme)
Pin / Tag / Fork / Archive	Chat "... " menu — pin keeps chats at top, <code>#tag</code> shown as chip, fork copies the whole conversation as <code>(fork)</code> , archive hides chats into a collapsible "Archived (n)" section

Usage Statistics (Admin)

Period-filterable dashboard (1 / 7 / 30 / 90 days): card row (requests, total/prompt/completion tokens, active users, guest sessions), a canvas bar chart of daily tokens (no chart library, re-themes with light/dark), top-10 bars by model and by user, plus **Purge guest chats now** and **Reset usage stats** (danger confirm modal). Endpoints: `GET /api/admin/stats?days=N` (1–365), `POST /api/admin/guest-purge-now` , `PUT /api/admin/reset-usage` .

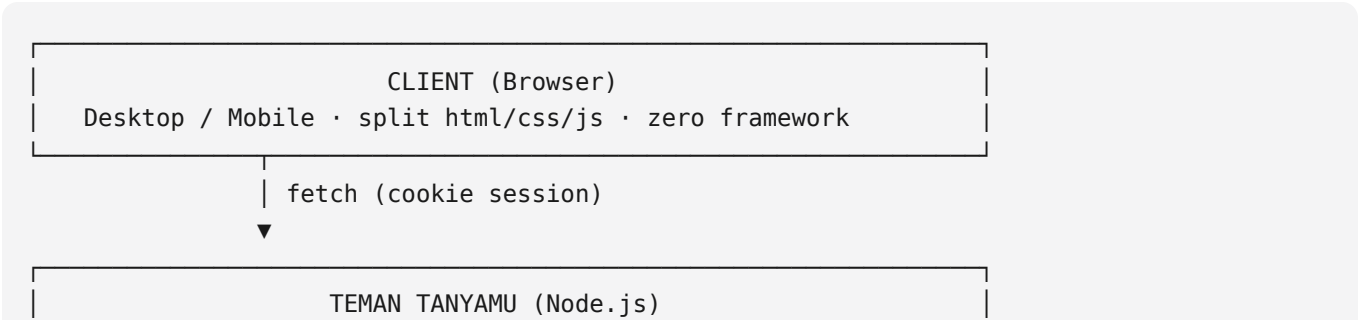
Models

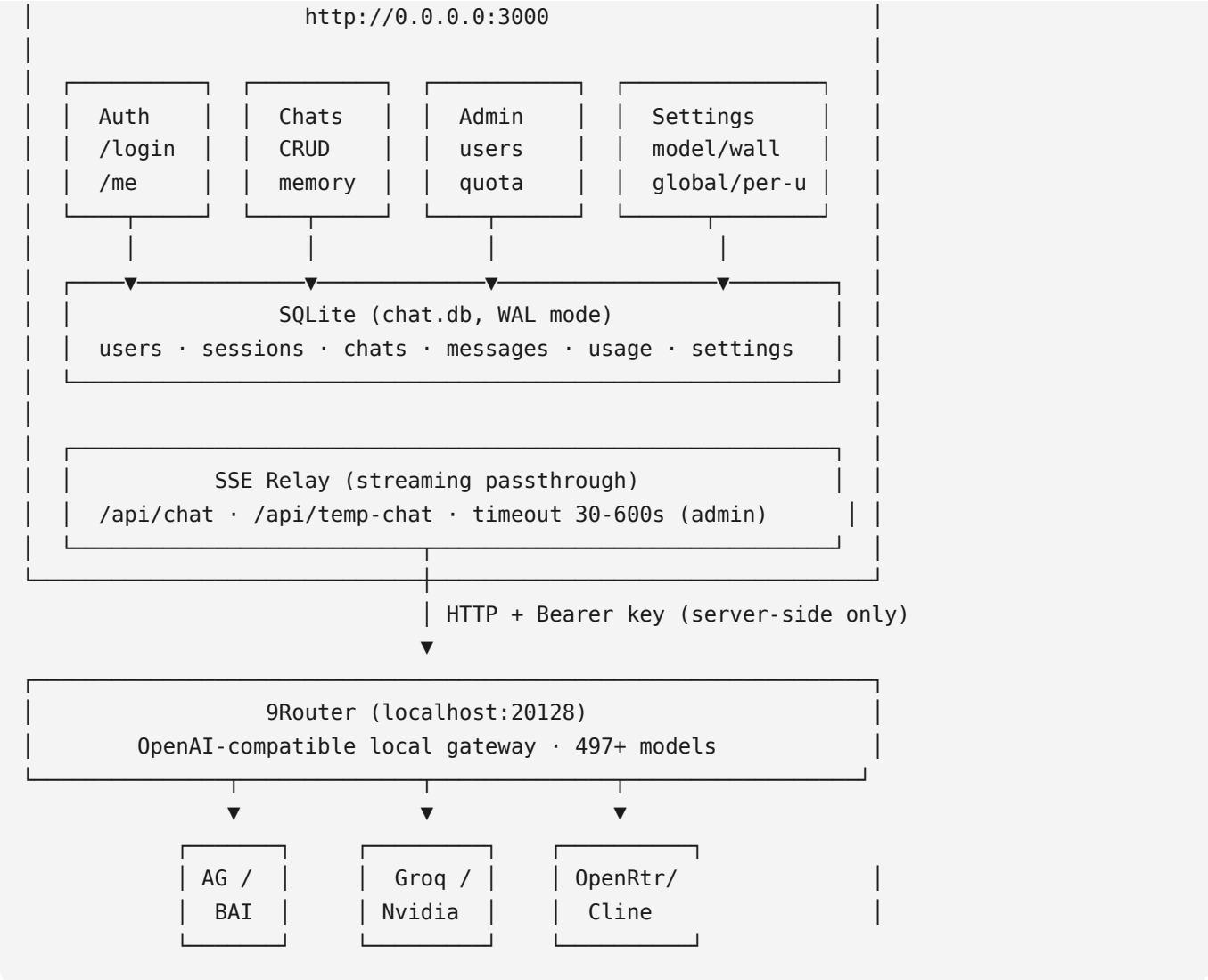
Feature	Description
Model Connection (gateway)	Base URL + API key set in Admin Dashboard (Open WebUI style): masked key display, show/hide eye, Test button (live latency + model count), instant hot-reload, no restart
Auto-fetch	Model list pulled live from 9Router (or any OpenAI-compatible endpoint)
Searchable picker	Custom combobox with instant filter + keyboard nav (↑/↓/Enter/Esc), works for admin & locked users
Global default	Admin setting, applies to all users
Per-user override	Admin sets specific model per user
Per-chat model	Each chat remembers its model
Locked for users	Regular users cannot change model (server-enforced); UI shows a red "Terkunci / Locked" chip

Appearance

Feature	Description
Light / dark theme	Toggle in the sidebar (sun/moon); dark = Open WebUI palette, light derived from it. Persisted per browser; highlight.js CSS theme swaps with it
Chat wallpaper	6 presets + custom image upload (per-user, optional global); dark overlay only for image wallpapers, gradient presets render as-is
Custom avatar	Auto-downscaled upload, strict base64 validation; rendered in chat bubbles, sidebar and admin list
Global assistant avatar	Admin uploads one AI avatar for everyone (/api/admin/assistant-avatar), or resets to default mask icon
Admin avatar popover	Click any avatar row in the admin user table → upload / reset / cancel (no native prompts anywhere)
Sidebar minimize	Desktop: icon rail (user-row collapses to avatar + stacked actions), state persisted · Mobile: closes the drawer
Bilingual UI	English / Indonesian, all strings translated — dialogs, toasts, admin panel, error codes
Custom dialogs	All confirm/rename/quota dialogs are in-app modals (i18n + consistent style), never native alert/confirm/prompt
Responsive	Mobile drawer sidebar, 44px+ tap targets, admin tables become card stacks, verified 1920→320 px with no horizontal overflow

System Architecture



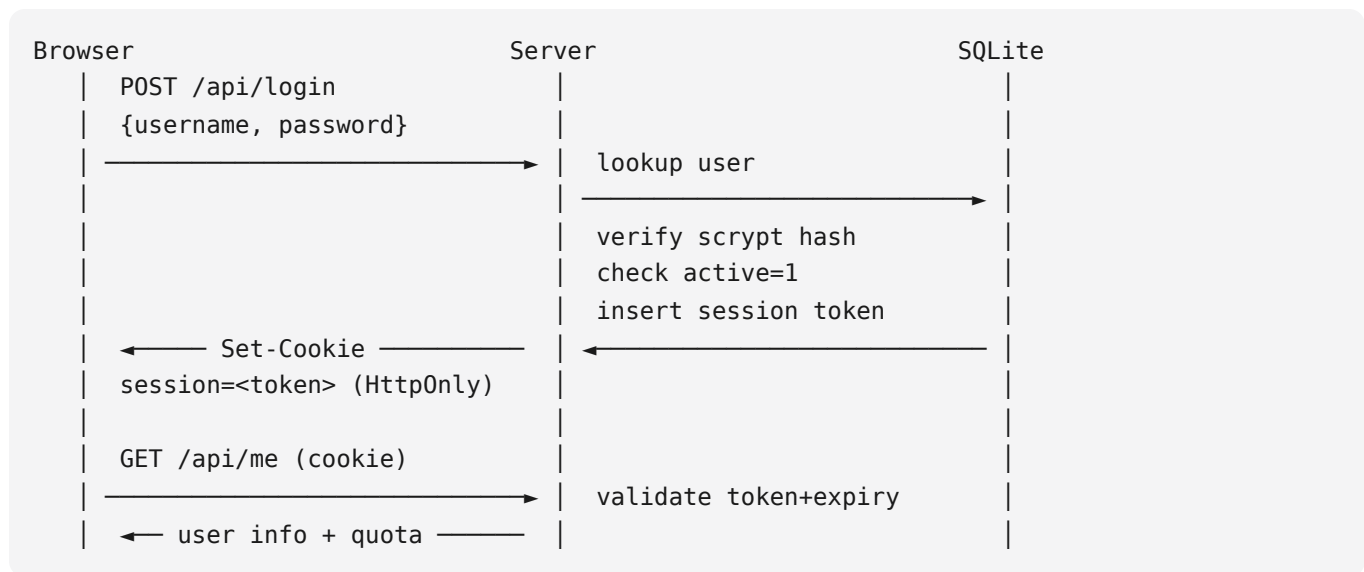


Key Design Decisions

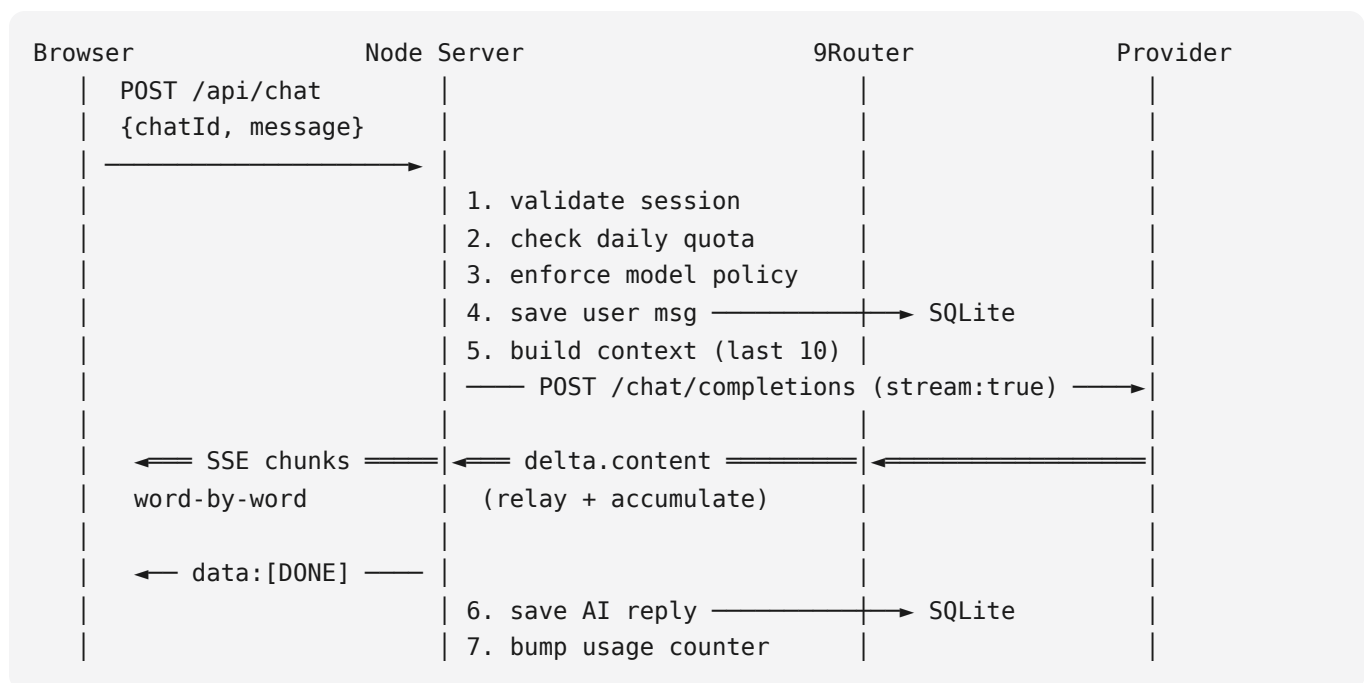
Decision	Reason
Single file backend (<code>server.js</code>)	Easy audit, zero complexity
SQLite over Postgres	Zero-config, WAL mode, single file backup
Server-side API key proxy	Key never reaches the browser
SSE relay (not direct)	Streaming works cross-device, CORS-free
Error codes (<code>ERR_*</code>)	Translated client-side per language
no-cache delivery + versioned assets	HTML: no-store · CSS/JS: no-cache + <code>?v=</code> query (updates always fresh, stale caches bust on release)
Split frontend (<code>style.css</code> + <code>app.js</code>)	Cacheable assets, lintable JS, CSP-ready

Request Flow Diagrams

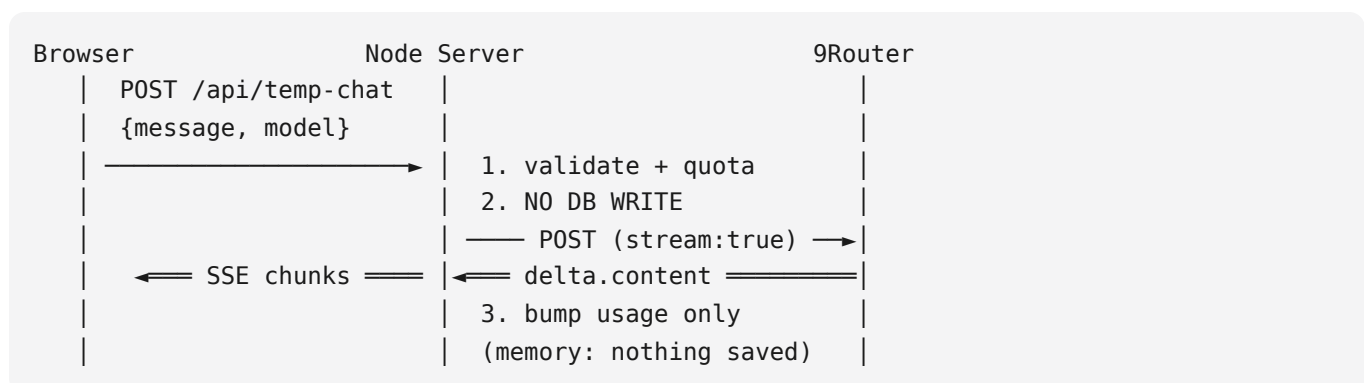
1. Login Flow



2. Streaming Chat Flow



3. Temporary Chat Flow



4. Admin Model Policy

User (non-admin) changes model?

- UI: selector disabled
- Server (double check):
 - PUT /api/me/model → 403 ERR_ADMIN_ONLY
 - PATCH /api/chats model → 403 ERR_ADMIN_ONLY
 - POST /api/chats {model} → forced to resolveUserModel()
 - POST /api/chat → chat.model auto-corrected

Admin sets global default → settings.default_model

Admin sets user override → users.model_override

Resolution order: user override > global default

Requirements

Component	Minimum
OS	Linux (tested: Ubuntu 24.04)
Node.js	>= 20 (tested: v24.21)
9Router	Running, OpenAI-compatible, default port 20128
Disk	~50 MB (incl. node_modules)

Installation

cd dash_ai_me
npm install # express + better-sqlite3
node server.js # http://localhost:3000

First run creates the admin account automatically:

username: admin
password: admin123 # change immediately via UI

Management Script (`manage.sh`)

All-in-one service manager (systemd user service):

```
./manage.sh start          # Start service
./manage.sh stop           # Stop service
./manage.sh restart        # Restart service
./manage.sh status         # Status, port, RAM
./manage.sh port <1-65535> # Change port (persistent, auto-restart)
./manage.sh clear-cache    # Clean SQLite WAL + npm cache + old journal
./manage.sh enable-boot    # Auto-start at boot (enables linger)
./manage.sh disable-boot   # Disable boot autostart
./manage.sh logs [n]       # Last n log lines (default 50)
./manage.sh password       # Reset admin password to admin123
./manage.sh help           # Show help
```

`enable-boot` runs `loginctl enable-linger` so the service runs without an active desktop session.

Configuration

Environment Variables

Variable	Default	Description
<code>PORT</code>	<code>3000</code>	Web server port
<code>ROUTER_BASE</code>	–	Boot fallback base URL (used only until set in Admin Dashboard)
<code>ROUTER_KEY</code>	–	Boot fallback API key (same; DB settings take precedence)
<code>ADMIN_PASSWORD</code>	<code>admin123</code>	Initial admin password (first run only)

Example:

```
PORT=8080 ROUTER_BASE=http://192.168.1.10:20128/v1 ROUTER_KEY=*** node server.js
```

Once set via **Admin Dashboard** → **Model Connection**, values in the `settings` table win over env vars — you can change gateway/key anytime from the browser without restart.

File Locations

Path	Description
<code>chat.db</code>	SQLite database (WAL mode)
<code>public/index.html</code>	Struktur halaman SPA (login + app shell)
<code>public/style.css</code>	Seluruh styling frontend
<code>public/app.js</code>	Seluruh logika frontend
<code>server.js</code>	Backend (single file)
<code>.port</code>	Persisted port number (created by <code>manage.sh port</code>)

Path	Description
<code>~/.config/systemd/user/dash-ai-me.service</code>	systemd unit

API Reference

All endpoints require the session cookie except `/api/login` and `/api/health`. Errors return `{"error": "ERR_*"}` codes, translated client-side.

Auth

Method	Endpoint	Body	Description
POST	<code>/api/login</code>	<code>{username, password}</code>	Sign in
POST	<code>/api/logout</code>	-	Destroy session (user + guest cookies)
GET	<code>/api/me</code>	-	User info + quota + effective model + global assistant avatar (guest: <code>guest</code> , <code>quota_used/max</code> , <code>seconds_left</code> , <code>expired</code>)
POST	<code>/api/guest</code>	-	Start/reuse guest session (cookie <code>gtoken</code> ; 403 when guest mode off)
GET	<code>/api/guest-config</code>	-	<code>{enabled}</code> — login screen show/hide for the guest button (no auth)
GET	<code>/api/health</code>	-	Liveness + DB check (no auth, for uptime monitors)

Account

Method	Endpoint	Body	Description
PUT	<code>/api/me/password</code>	<code>{current, next}</code>	Change own password
PUT	<code>/api/me/avatar</code>	<code>{avatar: dataURL null}</code>	Set/remove avatar
PUT	<code>/api/me/model</code>	<code>{model}</code>	Own model override (admin)
DELETE	<code>/api/me/model</code>	-	Clear override (admin)

Chats

Method	Endpoint	Body	Description
GET	<code>/api/chats</code>	-	List own chats
POST	<code>/api/chats</code>	<code>{title?, model?}</code>	Create chat
GET	<code>/api/chats/:id</code>	-	Chat + full messages
PATCH	<code>/api/chats/:id</code>	<code>{title?, model?, pinned?, archived?, tag?}</code>	Rename / lock model (admin) / pin / archive / tag

Method	Endpoint	Body	Description
POST	/api/chats/:id/fork	-	Copy chat + messages into a new (fork) chat
PUT	/api/chats/:id/messages/:msgId	{content}	Edit message content (user messages only)
DELETE	/api/chats/:id/messages/:msgId	-	Truncate chat: delete that message + everything after it
PATCH	/api/chats/:id/lean	{lean: bool}	Toggle lean (no-history) token saver per chat
DELETE	/api/chats/:id	-	Delete chat + messages

Memory (cross-chat, per user)

Method	Endpoint	Body	Description
GET	/api/memory	-	List own memory facts + enabled flag
PUT	/api/memory	{facts: string[]}	Replace full fact list (max 20, 240 chars each)
DELETE	/api/memory/:factId	-	Delete one fact
POST	/api/memory/clear	-	Wipe own memory

Chat & Preferences

Method	Endpoint	Body	Description
GET	/api/models	-	Model list from 9Router
POST	/api/chat	{chatId, message}	Streamed chat (SSE); token-budget context + system memory injected; final usage event with prompt/completion split
POST	/api/temp-chat	{message, model?, lean?}	Streamed temp chat (SSE, not saved); lean:true = zero context (no system prompt/memory)
GET	/api/wallpaper	-	Effective wallpaper
PUT	/api/wallpaper	{wallpaper null}	Own wallpaper

Admin

Method	Endpoint	Body	Description
GET	/api/admin/users	-	Users + today's usage + effective model + me id
POST	/api/admin/users	{username, password, role, daily_quota}	Create user
DELETE	/api/admin/users/:id	-	Delete user (self-delete blocked)
PUT	/api/admin/users/:id/password	{password}	Reset password (revokes sessions)
PUT	/api/admin/users/:id/active	{active: bool}	Activate / deactivate (self-deactivate blocked)

Method	Endpoint	Body	Description
PUT	/api/admin/users/:id/quota	{daily_quota}	Edit daily quota (1–999999)
PUT	/api/admin/users/:id/reset-quota	-	Reset one user's daily usage
PUT	/api/admin/reset-usage	-	Reset all users' usage (DELETE FROM usage)
PUT	/api/admin/users/:id/model	{model null}	Per-user model override
POST	/api/admin/avatar	{userId, avatar null}	Upload / reset any user's avatar (self-sync immediate)
POST	/api/admin/assistant-avatar	{avatar null}	Global assistant (AI) avatar for all users
GET	/api/admin/stats	?days=1..365 (def 30)	Usage statistics: totals, per-day series, top models/users, guest sessions
POST	/api/admin/guest-purge-now	-	Purge all guest data immediately
GET	/api/admin/settings	-	Global settings + today's token totals (prompt/completion)
PUT	/api/admin/settings	{default_model?, history_token_budget?, max_reply_tokens?, memory_enabled?, timeout_ms?, guest_enabled?, guest_max_chats?, guest_max_minutes?, guest_purge_hour?}	Update settings (budget 200–32000, reply cap 64–8192, timeout 30–600 s, guest chats 1–200, guest minutes 1–60, purge hour 0–23)
GET	/api/admin/router-config	-	Model gateway base URL + masked key + source (database / env)
PUT	/api/admin/router-config	{base_url?, api_key?}	Set gateway base URL & API key (hot-reload, no restart; empty key = keep)
POST	/api/admin/router-test	{base_url?, api_key?}	Probe GET {base}/models with saved or posted (pre-save) creds → latency + model count
PUT	/api/admin/wallpaper	{wallpaper null}	Set global wallpaper

Error Codes

Code	Meaning
ERR_UNAUTH	Not signed in

Code	Meaning
ERR_ADMIN_ONLY	Admin-only action
ERR_FIELDS_REQUIRED	Missing fields
ERR_LOGIN	Wrong credentials
ERR_INACTIVE	Account deactivated
ERR_PW_SHORT	Password under 4 chars
ERR_OLD_PW	Old password wrong
ERR_QUOTA	Daily quota exceeded
ERR_RATE_LIMITED	Too many failed logins: 5 attempts, then 2-minute IP block
ERR_NOT_FOUND	Resource not found
ERR_USERNAME_TAKEN	Username exists
ERR_TOO_BIG	Image too large
ERR_AVATAR_FORMAT	Avatar not an image
ERR_WALLPAPER_FORMAT	Wallpaper format invalid
ERR_SELF_DELETE	Cannot delete own account
ERR_SELF_DEACTIVATE	Cannot deactivate own account
ERR_BAD_BODY	Malformed JSON body
ERR_BAD_BUDGET	History budget out of range (200–32000)
ERR_BAD_MAXTOK	Reply token cap out of range (64–8192)
ERR_BAD_TIMEOUT	Upstream timeout out of range (30–600 s)
ERR_ROUTER_DOWN	9Router unreachable
ERR_ROUTER_NOT_CONFIGURED	Base URL / API key not set yet (Admin Dashboard → Model Connection)
ERR_BASE_REQUIRED	Base URL field empty
ERR_BASE_INVALID	Base URL not a valid http(s) URL
ERR_BASE_BLOCKED_HOST	Base URL host blocked (link-local / metadata addresses)
ERR_TIMEOUT	Upstream model call timed out
ERR_GUEST_DISABLED	Guest mode is turned off
ERR_GUEST_CHATS	Guest message limit reached
ERR_GUEST_TIME	Guest session time limit reached
ERR_GUEST_NOPE	Account feature unavailable in guest mode

Database Schema

```
users      (id, username, password_hash, role, daily_quota,
           model_override, avatar, active, created_at)
```

```

-- password_hash: "script1:<salt>:<hash>" (per-user salt);
-- legacy static-salt hex auto-upgrades on next login
sessions (token, user_id, expires_at) -- expired rows purged on login
chats (id, user_id, title, model, lean, pinned, archived, tag,
      summary, created_at, updated_at) -- lean: 1 = no-history token saver
-- pinned/archived: 0/1; tag: free text (<= 24 chars)
messages (id, chat_id, role, content, tokens, created_at)
usage (user_id, date, request_count, tokens_used) -- PK (user_id, date)
guest_sessions (token, guest_id, msgs_used, started_at, expires_at)
-- guest chats stored under user_id = -guest_id; wiped on logout / purge
memories (id, user_id, content, source, updated_at) -- cross-chat facts (max 20/user)
settings (key, value)
-- keys: default_model, wallpaper, assistant_avatar, router_base, router_key,
--       user:<id>:wallpaper, history_token_budget, max_reply_tokens,
--       memory_enabled, timeout_ms, memory_sync_chat, memory:last:<id>,
--       guest_enabled, guest_max_chats, guest_max_minutes, guest_purge_hour

```

Access from Other Devices

Server binds to all interfaces. From phones / LAN devices:

```
http://<laptop-ip>:3000      e.g. http://20.20.100.6:3000
```

For access over the internet, a Cloudflare Tunnel works out of the box: `trust proxy` is enabled and `cf-connecting-ip` is honored for login rate limiting; the session cookie gets the `Secure` flag automatically when the request arrives via HTTPS.

Performance

Metric	Measured
RAM idle	20-30 MB
RAM peak (streaming)	50-65 MB
CPU idle	~0%
Boot time	< 3 s
Streaming first token	~0.5-2 s (provider dependent)

Security Notes

- Passwords: script with per-user random salt, stored as `script1:<salt>:<hash>`, verified with `crypto.timingSafeEqual`; legacy static-salt hashes upgrade automatically on the next successful login

- Sessions: HttpOnly cookie, SameSite=Lax, 7-day expiry; expired rows purged on login
- **Secure cookie flag**: auto-enabled when behind HTTPS (Cloudflare Tunnel sets `X-Forwarded-Proto`)
- **Login rate limiting**: 5 failed attempts per IP, then a 2-minute block (`ERR_RATE_LIMITED`)
- `trust proxy` enabled: client IP correctly detected behind Cloudflare (`CF-Connecting-IP`)
- **Security headers** on every response: `X-Content-Type-Options: nosniff` , `X-Frame-Options: DENY` , `Referrer-Policy: no-referrer`
- **XSS hardening**: avatar & wallpaper validated server-side against strict formats (base64 image types / `linear-gradient(...)` / `#hex` only — no `svg+xml` , no attribute breakout); all user-supplied values escaped at render
- 9Router API key never sent to browser (server-side proxy only)
- Per-user chat isolation enforced in every SQL query
- Model policy enforced server-side (not just hidden in UI)
- Guest mode: chat/time limits enforced server-side **before** the AI call; guests blocked from all account endpoints (`ERR_GUEST_NOPE`); guest chats isolated under negative user ids and excluded from the `usage` stats table
- Quota checked before every AI call; failed/aborted streams do not consume quota
- Deactivate / password reset revokes all sessions instantly
- Caching: HTML `no-store` , CSS/JS `no-cache` + `?v=` asset version in `index.html` (bump on each release so old client caches invalidate automatically)

Troubleshooting

Problem	Fix
"9Router offline"	Verify gateway: <code>curl http://localhost:20128/v1/models -H "Authorization: Bearer ***"</code> — or click Test in Admin Dashboard → Model Connection
Moved gateway / new key	Admin Dashboard → Model Connection: set Base URL + API key, Test, Save. Applies instantly, no restart
Forgot admin password	<code>./manage.sh password</code>
Port already in use	<code>./manage.sh port 3001</code>
Not auto-starting at boot	<code>./manage.sh enable-boot</code> , verify <code>logincctl show-user \$USER grep Linger</code>
Stale UI after update	Hard refresh: <code>Ctrl+Shift+R</code> (HTML/CSS/JS are all no-cache; rarely needed)
Full reset (nuclear)	<code>./manage.sh stop && rm -f chat.db* && ./manage.sh start</code> — all data gone

Changelog

v1.0-beta (current)

- **Chat Experience** — smart auto-title after first reply, inline edit & regenerate (truncate + resend), highlight.js code blocks with copy button, pin / tag / fork / archive chat menu
- **Light / dark theme** — sidebar toggle, persisted per browser, hljs + stats chart re-theme live
- **Usage Statistics (admin)** — 1/7/30/90-day cards, canvas daily-token chart, top models/users bars, purge-now + reset-stats buttons
- **Scheduled guest purge** — wipes all guest data daily at an admin-set hour (`guest_purge_hour` , default 00:00) + manual "Purge guest chats now" button
- **Guest Mode** — login-free access via a "Try as guest" button; per-guest message limit (1–200) and session time limit (1–60 min) configurable in Admin Dashboard; live countdown chip; in-app login popup on chats/time exhaustion; server-enforced quota + endpoint lockdown; bilingual (EN/ID); guest data purged on logout
- **Model Connection in Admin Dashboard** — gateway Base URL + API key stored in DB (no more hardcoded), masked key + eye toggle, live Test button (latency + model count), hot-reload, SSRF guard on base URL
- Searchable model combobox, per-user quota edit, realtime quota UI (post-reply + 5 s admin poll), global assistant avatar, custom dialogs (confirm/rename/quota), red model-lock chip, minimized sidebar fixes, versioned asset cache-busting